

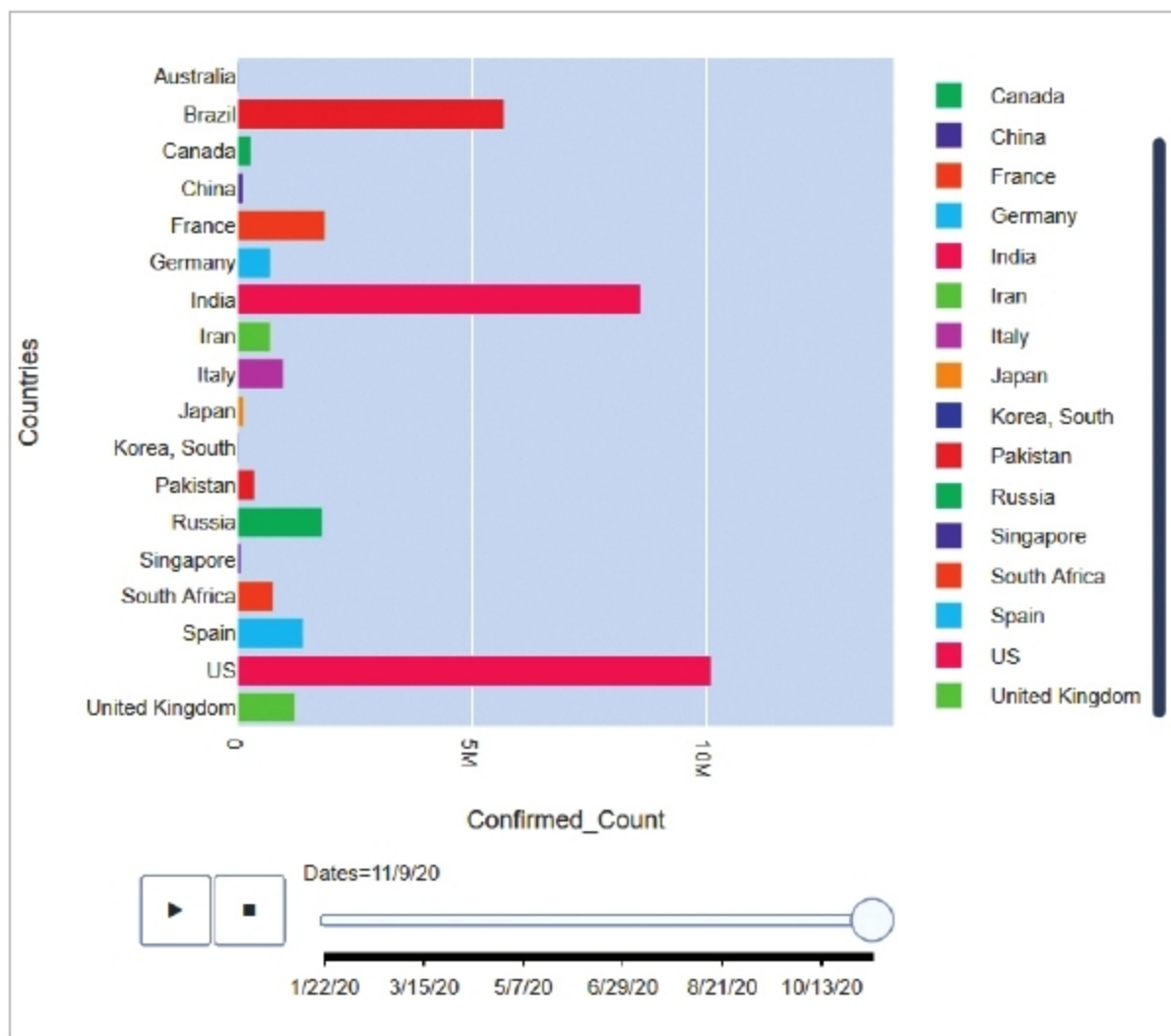


Figure 5: Interactive visualisation using Plotly

The output is shown in Figure 5.

Tensorflow, Pytorch, Keras: A brief introduction to the three main frameworks

TensorFlow, a product from Google, is an end-to-end machine learning library that helps to perform an extremely wide range of downstream tasks, with the primary focus on training and inference of deep neural networks. It is a symbolic math library based on dataflow and differentiable programming. TensorFlow bundles together machine learning and deep learning models and algorithms.

PyTorch is TensorFlow's direct competitor developed by Facebook, and is widely used in research projects. It allows almost unlimited customisation and is well adapted to running tensor operations on GPUs (similar to TensorFlow).

Keras is built on top of TensorFlow, which makes it a wrapper for deep learning purposes.

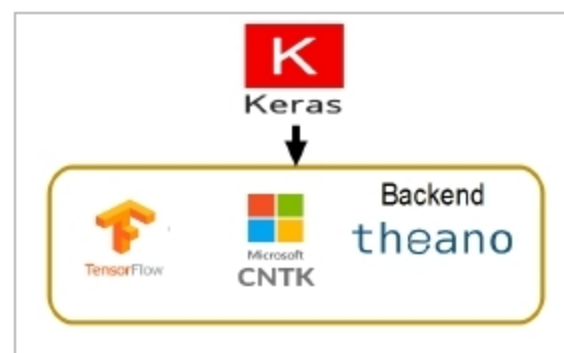


Figure 6: Keras, Tensorflow, CNTK, Theano relationship

It is incredibly user-friendly and simple to use. A solid asset is its

neural network block modularity and the fact that it is written in Python, which makes it easy to debug. Keras is backed by Google, Microsoft, Amazon, Apple and others. It is not a full-fledged deep learning framework, but is a nice wrapper around the libraries such as TensorFlow, CNTK, and Theano. Keras can be used directly; however, it is inbuilt with TensorFlow 2.0.

The following code is for where Keras is used directly:

```
import keras
keras.backend.backend('tensorflow')
from keras.models import Sequential
from keras.layers import Flatten, Dense, Activation
model = Sequential()
model.add(Flatten(input_shape=[28, 28]))
model.add(Dense(100, activation='relu'))
model.add(Dense(10, activation='softmax'))
```

The following code is used where Keras is inbuilt with TensorFlow 2.0:

```
from tensorflow import keras
model = keras.Sequential([
    keras.layers.Dense(10, input_shape=(784,)),
    activation= sigmoid
])
```

References

- [1] <https://datascience.foundation/sciencewhitepaper/data-science-brief-understanding-of-typical-project-life-cycle-tools-techniques-and-skills>
- [2] <https://datascience.foundation/sciencewhitepaper/natural-language-processing-nlp-simplified-a-step-by-step-guide>
- [3] <https://www.udemy.com/>
- [4] <https://www.springboard.com/>

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